

D495

Big Data Foundations

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A ride-share company with offices in 9 countries receives GPS pings from 400,000 drivers every 4 seconds and must price trips in real time. Its data team uses a popular open-source stack. Which big data characteristic does the every-4-seconds, real-time requirement mainly describe?

- A. Volume.
- B. Veracity.
- C. Velocity.
- D. Variety.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A global booking platform runs a distributed database across three regions. A network partition cuts one region off. The platform, which has 12 million users, must never let two customers book the same seat. Under CAP, what must it give up during the partition?

- A. Consistency.
- B. Partition tolerance.
- C. Nothing; well-designed systems always keep all three.
- D. Availability: some requests are refused until the partition heals.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

A social media app keeps showing like counts during a network partition, even if some users see slightly stale numbers. Which CAP choice is it making?

- A. ACID durability.
- B. Availability over partition tolerance.
- C. Consistency over availability.
- D. Availability over consistency.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A bank with 1.4 million customers is choosing a database for account transfers, where a debit and its matching credit must both succeed or both fail. The architect prefers vendors headquartered in Europe. Which property set is required?

- A. ACID transactions.
- B. A schema-on-read data lake.
- C. Any product from a European vendor.
- D. BASE with eventual consistency.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A fraud team at a payments firm processing 2 billion transactions a year needs to find rings of accounts linked through shared devices, addresses, and cards, several hops apart. Which NoSQL model fits best?

- A. A key-value store.
- B. A graph database.
- C. A wide-column store, because 2 billion rows demands it.
- D. A document store.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

Which workload is the best fit for a key-value store?

- A. Fetching a user's session data by session ID in milliseconds.
- B. Any workload, since key-value stores support every query type.
- C. Finding friend-of-friend connections.
- D. Ad-hoc joins across many tables for finance reports.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A media company wants to store raw clickstream JSON, video files, and sensor logs now and decide on structure later, when data scientists explore them. Storage is on a cloud provider it has used since 2019. Which architecture fits?

- A. An OLTP relational database.
- B. Whatever the provider set up in 2019.
- C. A data lake (schema-on-read).
- D. A data warehouse (schema-on-write).

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

Which description fits a data lakehouse?

- A. A raw lake with no metadata or governance.
- B. Low-cost lake storage with a table layer that adds ACID transactions, schema enforcement, and warehouse-style BI queries.
- C. A warehouse that stores only curated, structured data with schema-on-write.
- D. A replacement for all operational databases.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A retailer with 600 stores needs a job that totals each store's sales overnight for next-morning reports. Its point-of-sale terminals are five years old. Which processing approach fits?

- A. Stream processing, because sales arrive continuously.
- B. Batch processing on a nightly schedule.
- C. Replace the terminals before choosing.
- D. Always stream, because batch is obsolete.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

Which requirement forces stream processing rather than batch?

- A. Storing ten years of history.
- B. Producing a monthly revenue report.
- C. Retraining a model once a week.
- D. Blocking a suspicious card transaction within 200 milliseconds of the swipe.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A startup's single database server, which cost \$18,000, is at the largest CPU and memory the vendor offers and is maxed out. Traffic is expected to triple. What is the standard big data approach?

- A. Tune queries, which can always absorb triple traffic.
- B. Upgrade to a larger single server.
- C. Scale horizontally: partition the data across multiple nodes.
- D. Add a standby replica that takes over only on failure.

Target word _____	The one fact that decides _____	Answer ___	☐ sure ☐ unsure ☐ guess
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12.

A word-count job over 50 TB of log files runs on a 200-node cluster. The cluster was bought used from another firm. In MapReduce, what happens between the map phase and the reduce phase?

- A. Shuffle and sort: map outputs are grouped by key and sent to reducers.
- B. All raw data is copied to one central server.
- C. Reduce outputs are sent back to the mappers.
- D. Nothing; reducers read the raw files directly.

Target word _____	The one fact that decides _____	Answer ___	☐ sure ☐ unsure ☐ guess
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STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A ride-share company ~~with offices in 9 countries~~ receives GPS pings from 400,000 drivers every 4 seconds and must price trips in real time. ~~Its data team uses a popular open source stack.~~ Which big data characteristic does the every-4-seconds, real-time requirement mainly describe?

Target word: which V **Decides:** Speed of arrival and processing.

A	Adjacent category	Volume is total size; the question points to speed.
B	Confusion set	Veracity is data trustworthiness.
C	Correct	Velocity is the speed of data arrival and processing.
D	Confusion set	Variety is the range of data formats.

2. Answer D Concept 2

A global booking platform runs a distributed database across three regions. A network partition cuts one region off. The platform, ~~which has 12 million users~~, must never let two customers book the same seat. Under CAP, what must it give up during the partition?

Target word: property sacrificed **Decides:** Consistency is required (no double-booking).

A	Inverted operation	Consistency is what the business requires.
B	Confusion set	Partitions happen regardless; a distributed system must tolerate them.
C	Absolute language	CAP says this is impossible during a partition.
D	Correct	Keeping consistency during a partition means some nodes must stop answering.

3. Answer D Concept 2 · transfer

A social media app keeps showing like counts during a network partition, even if some users see slightly stale numbers. Which CAP choice is it making?

Target word: name the choice — reverse **Decides:** It keeps answering and tolerates stale data.

A	Adjacent category	A transaction property, not the CAP trade-off.
B	Confusion set	Partition tolerance is not optional here.
C	Inverted operation	That would refuse requests rather than show stale data.
D	Correct	It answers every request, accepting stale reads.

4. Answer A Concept 3

A bank ~~with 1.4 million customers~~ is choosing a database for account transfers, where a debit and its matching credit must both succeed or both fail. ~~The architect prefers vendors headquartered in Europe.~~ Which property set is required?

Target word: transaction guarantee **Decides:** All-or-nothing updates.

A	Correct	Atomicity guarantees both sides commit or neither does.
B	Adjacent category	A storage pattern for analytics, not transaction processing.
C	Camouflage	Vendor location does not provide transaction guarantees.
D	Confusion set	Eventual consistency can briefly show one side without the other.

5. Answer B Concept 4

A fraud team at a payments firm ~~processing 2 billion transactions a year~~ needs to find rings of accounts linked through shared devices, addresses, and cards, several hops apart. Which NoSQL model fits best?

Target word: NoSQL model **Decides:** Multi-hop relationships.

A	Adjacent category	Fast single-key lookups, no relationship traversal.
B	Correct	Graphs traverse relationships across many hops efficiently.
C	Camouflage	Size does not decide the model; the access pattern does.
D	Confusion set	Good for nested records, weak for many-hop links.

6. Answer A Concept 4 • transfer

Which workload is the best fit for a key-value store?

Target word: workload for a model — reverse **Decides:** Lookup by a single key.

A	Correct	Direct lookup by key is what key-value stores do best.
B	Absolute language	They support lookups by key, not rich queries.
C	Confusion set	A graph workload.
D	Adjacent category	A relational or warehouse workload.

7. Answer C Concept 5

A media company wants to store raw clickstream JSON, video files, and sensor logs now and decide on structure later, when data scientists explore them. ~~Storage is on a cloud provider it has used since 2019.~~ Which architecture fits?

Target word: storage architecture **Decides:** Raw, varied data with structure applied later.

A	Adjacent category	Built for transactions on structured data.
B	Camouflage	Provider history does not decide the architecture.
C	Correct	Lakes hold raw data of any type and apply structure at read time.
D	Inverted operation	Requires structure before loading.

8. Answer B Concept 5 • transfer

Which description fits a data lakehouse?

Target word: define the lakehouse — reverse **Decides:** Lake storage plus table features.

A	Dropped element	Drops the table layer that defines a lakehouse.
B	Correct	Combines lake flexibility with warehouse reliability.
C	Confusion set	That is a traditional warehouse.
D	Adjacent category	Lakehouses serve analytics, not transaction processing.

9. Answer B Concept 6

A retailer ~~with 600 stores~~ needs a job that totals each store's sales overnight for next-morning reports. ~~Its point-of-sale terminals are five years old.~~ Which processing approach fits?

Target word: processing mode **Decides:** The result is needed once, the next morning.

A	Confusion set	Arrival pattern does not matter; the result deadline does.
B	Correct	Scheduled processing of accumulated data meets a next-morning deadline.
C	Camouflage	Terminal age does not decide batch vs. stream.
D	Absolute language	Batch remains the right fit for scheduled reports.

10. Answer D Concept 6 · transfer

Which requirement forces stream processing rather than batch?

Target word: requirement that needs streaming — reverse **Decides:** Sub-second response to each event.

A	Adjacent category	Volume, not speed.
B	Inverted operation	A classic batch job.
C	Confusion set	Scheduled and periodic, so batch fits.
D	Correct	Only streaming acts on events that fast.

11. Answer C Concept 7

A startup's single database server, ~~which cost \$18,000,~~ is at the largest CPU and memory the vendor offers and is maxed out. Traffic is expected to triple. What is the standard big data approach?

Target word: scaling approach **Decides:** Vertical scaling has hit its ceiling.

A	Absolute language	Tuning helps, but cannot guarantee a threefold gain.
B	Inverted operation	No larger server exists; vertical scaling is exhausted.
C	Correct	Adding nodes grows capacity beyond one machine's limit.
D	Confusion set	Failover improves availability, not capacity.

12. Answer A Concept 8

A word-count job over 50 TB of log files runs on a 200-node cluster. ~~The cluster was bought used from another firm.~~ In MapReduce, what happens between the map phase and the reduce phase?

Target word: middle step **Decides:** Map output must be grouped by key before reducing.

A	Correct	Each reducer receives all values for its keys.
B	Inverted operation	MapReduce moves compute to the data, not data to one server.
C	Confusion set	Reverses the flow.
D	Dropped element	Drops the grouping step.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	2	
Wrong stem number	A figure copied from the prompt that is not the answer.	0	
Adjacent category	The right idea in the wrong bucket.	8	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	6	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	12	
Camouflage	Answers a stem fact that was never the question.	4	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	The Vs of big data	Volume = how much; velocity = how fast it arrives and must be processed; variety = how many forms; veracity = how trustworthy.	Calling every big-data challenge “volume.”	Time words (“every few seconds,” “real time”) vs. size vs. formats.
2	CAP theorem	During a network partition, a distributed system must choose between consistency and availability.	Believing a well-built system keeps all three during a partition.	“Must never double-book” (consistency) vs. “must always answer” (availability).
3	ACID vs. BASE	ACID transactions guarantee all-or-nothing, consistent updates; BASE systems accept eventual consistency for scale and availability.	Using an eventually consistent store for money movement.	“Both succeed or both fail.”
4	NoSQL data models	Key-value for fast lookups by key; document for flexible nested records; wide-column for massive sparse tables; graph for relationships across many hops.	Choosing by data size rather than by access pattern.	“Linked through,” “hops,” or “look up by ID.”
5	Warehouse, lake, lakehouse	Warehouses store curated, structured data with schema-on-write; lakes store raw data of any type with schema-on-read; lakehouses add table features (ACID, schema enforcement) on top of lake storage.	Putting raw, varied data into a warehouse, or calling a raw lake a lakehouse.	“Decide structure later” vs. “curated reports.”
6	Batch vs. stream processing	Batch processes accumulated data on a schedule; streaming processes events as they arrive within seconds or less.	Choosing streaming just because data arrives continuously, even when the need is a nightly report.	The deadline for the result: nightly vs. milliseconds.
7	Horizontal vs. vertical scaling	Vertical scaling buys a bigger machine and hits a ceiling; horizontal scaling adds nodes and partitions data across them.	Buying a bigger server after the largest size is already reached.	“Maxed out” or “largest the vendor offers.”
8	MapReduce flow	Map processes data where it lives and emits key-value pairs; shuffle-and-sort groups them by key; reduce aggregates each group.	Moving all data to one central server first.	“Between map and reduce.”

Course scope (public description): Big Data Foundations provides an in-depth introduction to big data concepts, terminology, and applications. You will learn the risks and challenges of working with extremely large data sets. The course introduces tools and techniques for working with big data. The course covers selection criteria for relational and non-relational data architectures and cloud-native data storage concepts. It also provides a historical perspective on the evolution of big data storage approaches. Data warehousing, data lakes, and data

lakehouses are introduced, and design principles for each are explained. Learners design aspects of big data architecture and big data processing to address given business requirements.